

■ Allows OEMs to add pre-boot features

Because the Framework is flexible, BIOS vendors can also use it to support pre-boot features for implementations: For example, BIOS vendors can add different types of application that run before the system boots. Consider what these could include:

■ A shell feature that can be used for a system test

■ A shell feature that boots from shell to Web browser, directly to an intranet support page, so that developers have instant access to information about solving problems while building the current configurations

■ A pre-boot Web browser that automatically goes to a port site and downloads the latest upgrade files for that server or system

In the future, the Framework could allow:

■ Self-configuration of systems using the Web

■ Call-home-after-crash capabilities

■ The instant-on feature

Phoenix's TrustedCore NB

Phoenix is not enthusiastic about EFI, and says it will wait until it becomes a standard. Meanwhile, Phoenix has its



Jargon Buster

Booting: The process of starting up a computer by loading the operating system. This takes place under the control of the boot loader program, usually located in a ROM chip.

Firmware: A term for software embedded in hardware. Such software cannot be modified.

CMOS: The part of a PC where permanent, un-erasable data resides.

Identity theft: The act of stealing someone's personal items, such as laptops and credit cards, and then masquerading as that person for personal gain.

Digital Rights Management: This refers to the control of who has access to what digital content. For example, when you download an e-book to read in Acrobat, Acrobat's DRM module checks whether you have the necessary permission to read that e-book.

Digital signature: A code used to verify that someone is indeed who he or she claims to be. This can be used in an encryption scheme to send and receive messages.

The Impending Change



The transition will be from a relatively dumb blue screen to a cleaner, friendlier environment, where virtually everything pre-boot can be managed effectively and more powerfully

own plans: The biggest maker of BIOSes has released the first entry in a line of next-generation products that it promises will:

■ Drive utility computing

■ Make PCs more secure

■ Prevent unauthorised users from misusing protected intellectual property

Phoenix believes that rather than serving just to boot the operating system, the BIOS of the future should do much more than just recognizing hardware components.

"The BIOS was designed for stand-alone devices", said Michael Goldgof, senior vice president of marketing for Phoenix. "Today everything is connected. Now there are new requirements. This is why created the Core System Software initiative."

In addition to trusted computing, Phoenix plans to build new features in the software, which is loaded when a computer system is turned on. CSS, which extends and replaces the BIOS we know today, will serve as a component for network, security, authentication and system management features.

The Content Management Engine (CME) TrustedCore NB for notebooks and tablet PCs is the first product in the CSS category, designed to extend the BIOS utility. TrustedCore NB is designed to allow businesses to keep their mobile computers safe from identity theft, unauthorised network access and unintentional data loss. Future versions hope to target servers, desktops and embedded systems such as consumer electronics, with plans to introduce digital rights management (DRM) and integrate the BIOS closer to Windows.

Phoenix's Core System Software (CSS) is a next-generation BIOS with a more sophisticated integration of OS and hardware—for example, it'll be easier for

system administrators to remotely monitor hardware configurations. It's built on a framework that Phoenix calls Device-Networked Architecture, or D-NA, and is part of a trend on the part of Microsoft and Intel to introduce 'trustworthy computing' into their products at a more basic level.

A cryptographic engine is inbuilt in the TrustedCore NB. It can be used to authenticate digital signatures and protect the Core System Software, while the BIOS allows manufacturers to create a protected area for the secure execution of inbuilt applications, which is protected from attacks by malicious code.

"Through our Core System Software, Phoenix is making a dramatic change that will become the basis of networked computing for the next two decades," said Phoenix chief executive, Chairman and President Albert E Sisto, in a statement. "Today, nearly all digital devices are connected to a network, whether to conduct global commerce, or just to access regular e-mail. This requires an advanced foundation for implementing an extensible and flexible architecture designed specifically for the age of networked computing."

Manufacturers will be able to protect system recovery and virus protection applications from malicious code. Self-management and self-authentication can be built into devices and servers at a basic level.

In a controversial turn, manufacturers will now be able to prevent users from tampering with areas of the system used for copy protection. And Phoenix recently said it is planning a BIOS with inbuilt DRM technology.

Phoenix has emphasised that the D-NA architecture is designed to fit neatly into Microsoft's plans for 'trustworthy computing', part of which is the contro-